

EARTHPRESS · REQUIREMENTS

EarthPress requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-10-02	Amish Chadha	R7 total set at 200 kg including the guard and bolts (EPR-DDR-003 A1, 2026-10-02), now met on paper; R5 status gives the softer design soil
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-10-01	Amish Chadha	Status from EPR-CAL-001 v0.3 for the constructable design (EPR-DDR-003); R7 and R13 not met, open decisions in EPR-DEC-001
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002): R5 reworded, R7 total relaxed to 190 kg, R13 at \$500; status from EPR-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	Status of every requirement from EPR-CAL-001; reference block and 2 MPa target adopted for TRL 3 (EPR-DDR-001); proposed revisions to R5, R7 and R13 listed, awaiting Amish
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates

0.1	2026-09-25	Amish Chadha	Initial scaffold
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EarthPress requirements

These requirements are checked by calculation in EPR-CAL-001 (TRL 3). On 2026-09-25 Amish accepted the TRL 3 recommendations (EPR-DDR-002), so R5, R7 and R13 carry the revised targets; the earlier targets are listed under "Revisions accepted" below. Targets still need validation with users in co-design sessions (see EPR-PRB-001).

The **reference block** is 290 x 140 x 90 mm, about 3.65 L and 6.94 kg dry, from a sandy clay loam stabilized with 5 % cement by dry mass (block size, 2 MPa target and stabilizer decided by Amish, 2026-09-25: go with recommendation; EPR-DDR-001 items 3 to 5, EPR-DDR-002).

Table 1. Requirements and status at TRL 3 (EPR-CAL-001 v0.2).

ID	Requirement	Target	Verification (TRL 3 or later)	Status (EPR-CAL-001)
R1	Block dimensions	290 x 140 mm $\pm$ 2 mm on plan; height 90 mm $\pm$ 3 mm, checked with the block gauge	Mold tolerance analysis; later gauge checks on a production run	Met on paper: 90.0 mm at the crank stop when the fill is within 0 to +2.7 %; mold bulge 0.08 mm
R2	Compaction pressure	2.0 MPa or more on the block face at the end of the stroke	Linkage force calculation	At risk: 81.2 kN reached with two operators (peak 668 N at the grip) and a weighed fill; one operator at 500 N stalls at 0.53 MPa
R3	Block strength with suitable soil	28-day dry compressive strength 4 MPa or more and wet strength 2 MPa or more (Auroville class B); in all cases 2.1 MPa (300 psi) or more, as in 14.7.4 NMAC	Literature review at TRL 3; block tests at TRL 4	Not verifiable at TRL 3
R4	Output	300 or more blocks in an 8 h day with a crew of four, including mixing	Cycle time estimate; later timed trials	At risk: 296 blocks a day at an assumed 85 s cycle
R5	Operator force	Peak pull of 500 N or less per operator at the lever grip; grip between 0.8 and 1.9 m above ground through the stroke, with the peak force at 1.0 m or higher	Force calculation; ergonomic check of the lever arc	Met on paper with two operators: 334 N each (668 N total) at 1.16 m for the base soil, and 405 N each (809 N total) for the softer soil on which the operating chart is based until partner soils are measured (2026-10-02); grip 0.86 to 1.89 m. Eject 470 N for one
R6	Garage-buildable	Built with a stick welder, angle grinder, drill press and hand tools; no turned or milled parts except purchased pins and bushings	Design review of every part	Met (design review): turned pins, shafts and bushes are bought
R7	Movable	Complete press 200 kg or less, including the guard and bolts (relaxed from 190 kg, EPR-DDR-003 A1, 2026-10-02); no single piece over 50 kg after removing pins and bolts; fits a 1.5 m pickup bed	Mass estimate from the model	Met on paper: 200 kg in total with the guard and bolts against the 200 kg total set on 2026-10-02; heaviest piece 45.0 kg, met; plan 1.28 x 0.58 m
R8	Durability	50,000 blocks before mold liners or bushings need replacing; all pins in replaceable bushings; grease points on every pivot	Wear estimate; later endurance log	Not verifiable at TRL 3: fatigue range 89 MPa against 243 MPa allowed; wear needs soil data
R9	Soil test kit	Field tests (jar sedimentation, shrinkage box, ribbon and drop tests) give a go or no-go and a starting cement content, with a first screen within 1 h and a full result within 24 h	Written procedure; later comparison with lab grading on partner soils	Met (design review); accuracy not verifiable at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (EPR-CAL-001)
R10	Low stabilizer use	Blocks reach R3 with 8 % cement by dry mass or less; 5 % as the design target	Literature at TRL 3; block tests at TRL 4	Not verifiable at TRL 3 (design at 5 %, 0.33 kg per block)
R11	Embodied carbon	Cradle-to-gate CO2 of the walling material 25 % or less of a fired-brick wall of equal volume	Carbon estimate from published factors	Met on paper: 7.6 % (0.35 t against 4.55 t for a small house)
R12	Safe operation	Lever cannot fall freely when released under load; lid latch holds against at least 1.5 times the peak block force; pinch points between lever and frame guarded or with 100 mm or more clearance	Design review; later proof-load test	Met on paper: rest catch and end pawl on the crank pin; lid hinge and latch pins checked at 173 kN (1.5 times is 121.8 kN); linkage guard modelled and clear of every moving part
R13	Affordable	Press and test kit parts at or below the value-engineering target of \$500, excluding labor and cement	Priced BOM (bom/bom.csv)	Over the value-engineering target by USD 20: \$520 (indicative) against the \$500 target (budget_usd , a hypothetical control target); savings to try are in EPR-DEC-001
R14	Open and documented	CERN-OHL-S drawings, BOM and build notes sufficient for a welder to build without contacting the author	Review by an outside fabricator at TRL 3	Not verifiable at TRL 3: no outside review yet

■ Requirements not met or at risk

Against the targets accepted on 2026-09-25, R7 was not met on paper once the design was made buildable on 2026-10-01 (EPR-DDR-003): 200 kg against 190 kg. On 2026-10-02 Amish set the R7 total at 200 kg including the guard and bolts and kept the 50 kg piece limit (EPR-DEC-001), so no requirement is now not met. One is over its value-engineering target: **R13** (estimated cost \$520, USD 20 over the \$500 target; the savings worth trying are in the register's Value engineering section). R2 and R4 remain at risk.

- **R2 at risk:** met only with two operators, a weighed fill (0 to +2.7 %) and the assumed soil stiffness; a softer soil (20 mm per decade) raises the peak to 809 N.
- **R4 at risk:** 296 against 300 blocks a day on an assumed cycle. Decided by Amish, 2026-09-25 (EPR-DDR-002): accept the 1 % shortfall and revisit when cycles are timed, which is TRL 4 work and on hold.
- **R3, R8, R10 and R14 cannot be verified at TRL 3.**

■ Revisions accepted (EPR-DDR-002)

Decided by Amish, 2026-09-25: go with recommendation. Applied in Table 1.

- **R5:** was a peak pull of 500 N or less with the grip between 0.9 and 1.6 m (not met: 668 N, and the band allows only 24.5° of arc). Now 500 N or less per operator, grip between 0.8 and 1.9 m, with the peak force at 1.0 m or higher.

- R7: was 140 kg or less in total (not met: 187 kg). Now 190 kg or less; the 50 kg piece limit and pickup bed are unchanged.
- R13: was \$450 or less (not met: \$488). Now \$500 or less, matching `budget_usd` in `project.yaml`.

■ Assumptions

- Compaction pressure of 2 to 4 MPa and about 5 % cement follow Auroville Earth Institute practice ([Auroville Earth Institute](#)).
- Dry block density about 1,900 kg/m³ (estimate), giving about 6.9 kg per dry block.
- One operator can pull 500 N at the grip for short strokes, using body weight; two can pull 1,000 N on the T-handle. To be checked with users.
- Soil pressure rises tenfold over the last 14 mm of the stroke (range 10 to 20 mm); to be measured with partner soils.
- A crew of four covers soil sieving, mixing, pressing and stacking.